

- 1a) i) $x_n = 0, 3, 6, 9, 12, \dots$: $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto 3n$
 ii) $x_n = -4, -1, 2, 5, 8, \dots$ $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto -4 + 3n$
 iii) $x_n = 0, -1, 2, -3, 4, \dots$ $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto (-1)^n \cdot n$
 iv) $x_n = 0, 1, -2, 3, -4, \dots$ $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto (-1)^{n+1} \cdot n$
 v) $x_n = 1/2, 1/4, 1/8, 1/16, \dots$ $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto \frac{1}{2^{n+1}}$
 vi) $x_n = 1, 1/2, 1/4, 1/8, 1/16, \dots$ $f: \mathbb{N} \rightarrow \mathbb{R}, n \mapsto \frac{1}{2^{n+1}}$

1b) $a_{n+1} = \frac{2a_n}{2+a_n}, n \geq 0, a_0 := 2$
 $a_1 = \frac{2a_0}{2+a_0} = \frac{2 \cdot 2}{2+2} = 1$
 $a_2 = \frac{2a_1}{2+a_1} = \frac{2 \cdot 1}{2+1} = 2/3$
 $a_3 = \frac{2a_2}{2+a_2} = \frac{2 \cdot 2/3}{2+2/3} = \frac{4/3}{8/3} = 1/2$

2) a) $x_n = n, y_n = -n \Rightarrow x_n + y_n = 0$

$x_n = (-1)^n, y_n = (-1)^{n+1} \Rightarrow x_n + y_n = 0$

b) $x_n = 0, y_n = (-1)^n \Rightarrow x_n \cdot y_n = 0$

$x_n = n, y_n = 1/n \Rightarrow x_n \cdot y_n = 1$

c) $x_n = 0, y_n = 1/n$

$\lim_{n \rightarrow \infty} x_n = 0 = \lim_{n \rightarrow \infty} y_n$

d) $x_n = -1^n, |x_n| = 1$

$$3) a) x_n = \frac{3n^2 + 4n + 20}{4n^3 + 1000} = \frac{n^2 \cdot (3 + \frac{4}{n} + \frac{20}{n^2})}{n^3 \cdot (4 + \frac{1000}{n^3})} = \frac{1}{n} \cdot \frac{3 + \frac{4}{n} + \frac{20}{n^2}}{4 + \frac{1000}{n^3}} = \frac{1}{n} \cdot \frac{3}{4} = 0 \Rightarrow \lim_{n \rightarrow \infty} x_n = 0$$

$$b) x_n = \frac{2n^3 + 7n^2 + 12}{-5n^3 - n + 3} = \frac{n^3 (2 + \frac{7}{n} + \frac{12}{n^3})}{n^3 (-5 - \frac{1}{n^2} + \frac{3}{n^3})} \Rightarrow \lim_{n \rightarrow \infty} x_n = -\frac{2}{5}$$

$$c) x_n = (2 + \frac{3}{n})^5 \approx (2 + 0)^5 \Rightarrow \lim_{n \rightarrow \infty} x_n = 2^5 = 32$$

$$d) x_n = \sqrt[2n]{2^{1000}} = 2^{\frac{1000}{2n}} = 2^{\frac{500}{n}} = \sqrt[n]{2^{500}} \Rightarrow \lim_{n \rightarrow \infty} x_n = 1 \quad \lim_{x \rightarrow \infty} \sqrt[x]{x} = 1$$

$$e) x_n = \left(\frac{n+1}{n}\right)^n = \left(1 + \frac{1}{n}\right)^n \Rightarrow \lim_{n \rightarrow \infty} x_n = e^1 \quad \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$f) x_n = (-1)^n \cdot \frac{\sin(n) \cdot \cos(n)}{3n^3}, \text{ da } \sin(n) \text{ und } \cos(n) \in [-1; 1] : \lim_{n \rightarrow \infty} x_n = 0$$

$$4) a) \left|\frac{1}{n} - 0\right| < \varepsilon \text{ gilt, sofern } n > \frac{1}{\varepsilon^2}$$

$$b) \text{ Behauptung } \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

ε	10	1	1/10	1/100
$1/\varepsilon^2$	1/100	1	100	10 ⁴
N	1	2	101	10 ⁴ +1

Sei $\varepsilon > 0$ und $N > 1/\varepsilon^2$ mit $N \in \mathbb{N}$

$$\Rightarrow n \geq N : \left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \frac{1}{\sqrt{N^2}} = \varepsilon$$

Somit gilt $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

$$5) a) \sum_{k=0}^5 \left(\frac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^6}{1 - \frac{1}{2}} = \frac{1 - 1/64}{1/2} = \frac{63/64}{1/2} = 63/32$$

$$b) \sum_{k=1}^5 \left(\frac{1}{2}\right)^k = \sum_{k=0}^5 \left(\frac{1}{2}\right)^k - \left(\frac{1}{2}\right)^0 = 63/32 - 1 = 31/32$$

$$c) \sum_{k=2}^5 \left(\frac{1}{2}\right)^k = \sum_{k=0}^5 \left(\frac{1}{2}\right)^k - \left(\left(\frac{1}{2}\right)^0 + \left(\frac{1}{2}\right)^1\right) = 63/32 - 3/2 = 15/32$$

$$d) \sum_{k=0}^{10} (-1)^k = \frac{1 - (-1)^{11}}{1 - (-1)} = \frac{1 - (-1)}{2} = 1$$

$$e) \sum_{k=0}^{11} (-1)^k = \frac{1 - (-1)^{12}}{1 - (-1)} = \frac{0}{2} = 0$$

$$f) \sum_{k=0}^2 3^k = 3^0 + 3^1 + 3^2 = \frac{1 - 3^3}{1 - 3} = \frac{1 - 27}{-2} = 13$$